Special Numbers

Year 4 Pt 2 · **Mark Scheme**

Lesson Worksheet — Answers

Square & Cube Basics

1. 1, 4, 9, 16, 25, 36

2. 49

3. 7

4. C) 8

5. 3

6. Yes — 6 × 6 = 36, so 36 is a square number.

7. 1, 3, 6, 10

8. C) 15

9. 100

10. D) 64

11. A, B and D (9, 16 and 25 are square numbers; 20 and 32 are not)

12. 10

13. 64

14. A, C and E (6, 10 and 15 are triangular numbers; 8 and 14 are not)

15. C) 17 ($3^2 = 9$, $2^3 = 8$, and $9 + 8 = 17$)

Patterns & Classifying

16. 64 ($8^2 = 64$, which is even and lies between 50 and 100)

17. 36 (the sum of the first n odd numbers is n^2, so the sum of the first 6 odd numbers is $6^2 = 36$)

18. 28 (1+2+3+4+5+6+7 = 28)

19. A) 21 (1+2+3+4+5+6 = 21)

20. A, B, C and D are true (E is false — 64 is an even number, not odd)

21. 10 and 15 (the 4th and 5th triangular numbers: 10 + 15 = 25)

22. 11 ($6^2 - 5^2 = 36 - 25 = 11$)

23. D) 64 (27 is below 30, and 64 is the only cube number between 30 and 100)

24. 45 (the gap between terms grows by 1 each time; the gap to reach the 8th term was +8, so the gap to the 9th term is +9, giving 36 + 9 = 45)

25. A and D (1 and 36 are both square numbers and triangular numbers)

26. 25 (1+3+5+7+9 = 25); this is a square number (5^2)

27. D) 81 ($\sqrt{81} = 9$, which is odd and greater than 8)

28. 21 (1+2+3+4+5+6 = 21 tins)

29. C) 16 ($5^2 - 3^2 = 25 - 9 = 16$)

30. Example: the 3rd triangular number is 6 and the 4th is 10; 6 + 10 = 16 = 4^2. In general, two copies of a triangular arrangement of side n and side (n−1) fit together to form a square of side n, so T(n) + T(n−1) = n^2.

Special Numbers Challenge

31. 169 ($13^2 = 169$; $12^2 = 144$ is too small)

32. 125 ($5^3 = 125$; $6^3 = 216$ is too big)

33. C) 16 ($\sqrt{16} = 4$, and 4 is itself a square number greater than 1 — this is the smallest square number with this property)

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34. 66 (the 11th triangular number = 55 + 11 = 66)

35. C and D (28 and 36 are even triangular numbers between 10 and 50; 15, 21 and 45 are odd triangular numbers)

36. $7^2 + 7 = 49 + 7 = 56$. The 7th triangular number is $7 \times 8/2 = 28$, and $2 \times 28 = 56$, confirming the identity.

37. 13 m ($\sqrt{169} = 13$)

38. E) 81 (a number that is one more than a multiple of 10 always ends in 1; 81 = 80 + 1)

39. 28 and 36 (the 7th triangular number is 21 + 7 = 28, and the 8th is 28 + 8 = 36)

40. C) 9 ($9^3 = 729$)

41. B, C and E are true (A is false — e.g. 8 is a cube number but not a square number; D is false — e.g. 9 is a square number but odd)

42. 36 (1+2+3+4+5+6+7+8 = 36 oranges)

43. C) 15 ($15^2 = 225$)

44. 36 (36 is both the 8th triangular number — 1+2+…+8 = 36 — and 6^2)

45. 100 (the sum of the first 10 odd numbers is $10^2 = 100$)

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Homework Worksheet — Answers

Square & Cube Basics

1. 1, 8, 27, 64, 125, 216

2. 81

3. 8

4. D) 27

5. 2

6. No — $7^2 = 49$ and $8^2 = 64$, and 50 lies between them, so 50 is not a square number.

7. 1, 3, 6, 10, 15

8. C) 21

9. 144

10. A) 27

11. A, C and E (4, 36 and 49 are square numbers; 12 and 44 are not)

12. 11

13. 125

14. A, C and E (3, 10 and 21 are triangular numbers; 5 and 12 are not)

15. C) 24 ($4^2 = 16$, $2^3 = 8$, and $16 + 8 = 24$)

Patterns & Classifying

16. 25 ($5^2 = 25$, which is odd and lies between 20 and 40)

17. 49 (the sum of the first n odd numbers is n^2, so the sum of the first 7 odd numbers is $7^2 = 49$)

18. 45 (1+2+3+4+5+6+7+8+9 = 45)

19. C) 28 (1+2+3+4+5+6+7 = 28)

20. A and C are true (B is false — 100 is not a cube number; D is false — 99 is not a cube number; E is false — 100 is even)

21. 6 and 10 (the 3rd and 4th triangular numbers: 6 + 10 = 16)

22. 9 ($5^2 - 4^2 = 25 - 16 = 9$)

23. E) 125 (64 is below 100, and 125 is the only cube number between 100 and 150)

24. 36 (the gap between terms grows by 1 each time; the gap to reach the 7th term was +7, so the gap to the 8th term is +8, giving 28 + 8 = 36)

25. A and E (1 and 36 are both square numbers and triangular numbers)

26. 36 (1+3+5+7+9+11 = 36); this is a square number (6^2)

27. D) 100 ($\sqrt{100} = 10$, which is even and greater than 8)

28. 28 (1+2+3+4+5+6+7 = 28 cans)

29. C) 20 ($6^2 - 4^2 = 36 - 16 = 20$)

30. Example: the 4th triangular number is 10 and the 5th is 15; $10 + 15 = 25 = 5^2$. In general, $T(n) + T(n-1) = n^2$, since two triangular arrangements of sides n and (n−1) fit together to form a square of side n.

Standard / Advanced

31. 225 ($15^2 = 225$; $14^2 = 196$ is too small)

32. 64 ($4^3 = 64$; $5^3 = 125$ is too big)

33. E) 81 ($\sqrt{81} = 9$, and 9 is itself a square number greater than 1)

34. 91 (the 13th triangular number = 78 + 13 = 91)

35. B, C and E are true (A and D are false — 10 and 28 are even triangular numbers, not odd)

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36. $8^2 + 8 = 64 + 8 = 72$. The 8th triangular number is $8 \times 9/2 = 36$, and $2 \times 36 = 72$, confirming the identity.

37. 14 m ($\sqrt{196} = 14$)

38. D) 49 (a number that is one less than a multiple of 10 always ends in 9; 49 ends in 9)

39. 55 and 66 (the 10th triangular number is 45 + 10 = 55, and the 11th is 55 + 11 = 66)

40. C) 8 ($8^3 = 512$)

41. A, B, D and E are true (C is false — 125 is a cube number but not a square number, since no whole number squared equals 125)

42. 45 (1+2+3+4+5+6+7+8+9 = 45 boxes)

43. C) 14 ($14^2 = 196$)

44. 36 (36 is both the 8th triangular number — 1+2+…+8 = 36 — and 6^2)

45. 144 (the sum of the first 12 odd numbers is $12^2 = 144$)

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